

AIPI 561 — Week 2: Deployment Screenshots

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1. kubectl get pods — two demand-api pods Running

Deliverable: kubectl showing pods running

```
arnavmahale@Arnavs-MBP ops-ai-arnavmahale-week2 % EXTERNAL_IP=34.136.184.3
arnavmahale@Arnavs-MBP ops-ai-arnavmahale-week2 % kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
demand-api-548897b7b5-r6vs4        1/1     Running   0           2m55s
demand-api-548897b7b5-z87lp        1/1     Running   0           2m56s
```

2. kubectl get svc demand-api — LoadBalancer with EXTERNAL-IP 34.136.184.3

Deliverable: kubectl showing external IP assigned

```
arnavmahale@Arnavs-MBP ops-ai-arnavmahale-week2 % kubectl get svc demand-api
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
demand-api	LoadBalancer	34.118.232.140	34.136.184.3	80:30495/TCP	14m

3. curl http://\$EXTERNAL_IP/health → {"status":"ok"}

Deliverable: API health check

```
arnavmahale@Arnavs-MBP ops-ai-arnavmahale-week2 % curl "http://$EXTERNAL_IP/health"
{"status":"ok"}%
```

4. curl /api/recommendations?zone_id=68&hour=17&dow=4 — ranked pickup zones JSON

Deliverable: at least one working API endpoint

```
arnavmahale@Arnavs-MBP: ops-a1-arnavmahale-week2 % curl "http://$EXTERNAL_IP/api/recommendations?zone_id=68&hour=17&dow=4&date=2026-01-15&n=36&holiday=regular"
[{"rank":1,"zone_id":70,"name":"East Elmhurst","borough":"Queens","trips":6.2,"demand_score":6,"drive_minutes":3,"est_yield_min":40,"est_yield_max":100,"efficiency_score":6.98336384439392}, {"rank":2,"zone_id":88,"name":"Financial District South","borough":"Manhattan","trips":15.6,"demand_score":15,"drive_minutes":3,"est_yield_min":24,"est_yield_max":99,"efficiency_score":4.39440168535117}, {"rank":3,"zone_id":87,"name":"Financial District North","borough":"Manhattan","trips":10.8,"demand_score":10,"drive_minutes":3,"est_yield_min":22,"est_yield_max":90,"efficiency_score":3.9223719538445}]
```

5. GitHub Actions — both CI and CD workflows green on push to main

Deliverable: working CI/CD pipeline (overview)

arnavmahale / ops-ai-arnavmahale-week2

Q Type to search

+

<> Code

Issues

Pull requests

Actions

Projects

Wiki

Security and quality

Insights

Settings

Actions

New workflow

All workflows

CD — Deploy to GKE

CI — Test & Build

Management

Caches

Attestations

Runners

Usage metrics

Performance metrics

All workflows

Showing runs from all workflows

Filter workflow runs

Help us improve GitHub Actions

Tell us how to make GitHub Actions work better for you with three quick questions.

Give feedback

2 workflow runs

Workflow

Event

Status

Branch

Actor

Week 2: deploy taxi demand API to GKE with CI/CD

CD — Deploy to GKE #1: Commit 6ea38d8 pushed by arnavmahale

main

4 minutes ago

3m 39s

Week 2: deploy taxi demand API to GKE with CI/CD

CI — Test & Build #1: Commit 6ea38d8 pushed by arnavmahale

main

4 minutes ago

1m 6s

6. CD workflow detail — all 17 steps passed (auth → build → push → set image → rollout)

Deliverable: working CI/CD pipeline (detail)

← CD — Deploy to GKE

🟢 Week 2: deploy taxi demand API to GKE with CI/CD #1

Re-run all jobs

Summary

All jobs

🟢 deploy

Run details

Usage

Workflow file

Annotations

1 warning

deploy

succeeded 4 minutes ago in 3m 36s

Search logs

> Set up job

2s

> Checkout code

1s

> Authenticate to Google Cloud

8s

> Set up Cloud SDK

21s

> Install gke-gcloud-auth-plugin

31s

> Get GKE cluster credentials

1s

> Configure Docker for Artifact Registry

1s

> Set up Docker Buildx

6s

> Build and push Docker image

1m 1s

> Update Deployment image

3s

> Wait for rollout

1m 24s

> Verify deployment

8s

> Post Build and push Docker image

1s

> Post Set up Docker Buildx

1s

> Post Authenticate to Google Cloud

8s

> Post Checkout code

8s

> Complete job

8s

7. GKE cluster + Artifact Registry deletion confirmation

Deliverable: GKE cluster deletion (cleanup); GCS bucket retained for Week 3

```
=== Step 1/3: Delete GKE cluster (this stops the meter - ~3-5 min) ===
Deleting cluster operationalizing-ai...done.
Deleted [https://container.googleapis.com/v1/projects/ops-ai-arnavmahale/zones/us-central1-a/clusters/operationalizing-ai].

=== Step 2/3: Confirm cluster is gone (this output is your deletion screenshot) ===
(if the table above is empty, deletion succeeded)

=== Step 3/3: Delete Artifact Registry repo ===
Delete request issued for: [docker-repo]
Waiting for operation [projects/ops-ai-arnavmahale/locations/us-central1/operations/0ad4da7f-444a-4323-b62d-423f0d079bba] to complete...done.
Deleted repository [docker-repo].

=== Confirming GCS bucket is RETAINED (needed for Week 3) ===
gs://ops-ai-arnavmahale-data/demand_enriched.parquet
gs://ops-ai-arnavmahale-data/lgbm_demand_model.txt
gs://ops-ai-arnavmahale-data/taxi_zones.geojson
gs://ops-ai-arnavmahale-data/zone_hour_avg_fare.parquet

=== Cleanup complete. ===
```